WORKED SOLUTION

STEP II 1994, Question 1

Pure • Divisibility and Decimal Digits

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q1

The key idea

The last few digits of a product depend only on the last few digits of the factors, so everything here is arithmetic modulo a power of 10. A number ending in 1, 3, 7 or 9 shares no factor with 10, and that makes it *invertible* modulo 10, and in fact modulo any power of 10. The middle part strips the factors of 2 and 5 out of n and turns them into zeros. The final part uses invertibility modulo 10^{10} to force the last ten digits of the product to be any block we like.

Solution

Part 1: a multiple ending in 1

The last digit of $n \times m$ is the last digit of (last digit of n)$\times$(last digit of m), because the other digits contribute multiples of 10. Since n doesn't end in 5 or an even digit, it ends in 1, 3, 7 or 9, and

$$1 \times 1 = 1, \qquad 3 \times 7 = 21, \qquad 7 \times 3 = 21, \qquad 9 \times 9 = 81.$$

So take $m = 1$, 7, 3 or 9 respectively. Then $n \times m$ ends in 1, as required.

Part 2: any n

Write

$$n = 2^a \, 5^b \, k,$$

where $a, b \geq 0$ and k is divisible by neither 2 nor 5. Then k doesn't end in 5 or an even digit, so by Part 1 there is m' with km' ending in 1. Now let

$$m = 2^b \, 5^a \, m'.$$

Then

$$n \times m = 2^{a+b} \, 5^{a+b} \, km' = 10^{a+b} \, (km').$$

Multiplying by 10^{a+b} just appends $a+b$ zeros to km'. So $n \times m$ ends in 1 if $a+b=0$, and in 1 followed by $a+b$ zeros otherwise. This is what we needed.

For example, $n = 40 = 2^3 \cdot 5$ has $k=1$, so $m = 2 \cdot 5^3 = 250$ and $n \times m = 10\,000$.

Part 3: all ten digits

Write $n = 10^r q$ with $r \geq 0$ and q ending in 1. Appending zeros never removes digits, so it is enough to find m with $q \times m$ containing every digit; then $n \times m = 10^r(qm)$ contains them too.

Let $N = 10^{10}$ and $X = 1234567890$, which contains every digit. We aim for

$$qm \equiv X \pmod{N},$$

that is, qm ends in the ten digits 1234567890.

> **Ask yourself** Why must some multiple of q leave remainder X on division by 10^{10}?

Consider the N numbers $q \times 0,\ q \times 1,\ \ldots,\ q \times (N-1)$ and their remainders modulo N. Suppose two were equal, say $qs \equiv qt \pmod{N}$ with $0 \leq t < s \leq N-1$. Then 10^{10} divides $q(s-t)$. But q ends in 1, so it is odd and not a multiple of 5, and it has no prime factor in common with $10^{10} = 2^{10}5^{10}$. Hence $2^{10}5^{10}$ divides $s-t$, which is impossible because $0 < s-t < N$. So the N remainders are all different. There are only N possible remainders, $0, 1, \ldots, N-1$, so every one of them occurs. In particular, some m with $0 \leq m \leq N-1$ has $qm \equiv X \pmod{N}$.

This m isn't 0, since $X \not\equiv 0$. So qm is a positive multiple with $qm \geq X \geq 10^9$. Its last ten digits are therefore exactly $1, 2, 3, 4, 5, 6, 7, 8, 9, 0$, with the leading 1 of X genuinely present. Thus $q \times m$, and hence $n \times m$, contains all the digits $0, 1, \ldots, 9$.

For instance, with $n = 21$ we can take $m = 7\,201\,646\,090$, and then $21 \times m = 151\,234\,567\,890$.

> **Ask yourself** Where was the hypothesis on n used, and why was it needed?

We used it only to know that q has no factor 2 or 5. Without that, the remainders qm mod 10^{10} can't take every value. For example, every multiple of 2 is even, so it can't end in $\ldots 1$. The hypothesis is what lets us dictate the *end* of the product. (Part 2 shows that any n has a multiple of this form, so combined with Part 3, *every* n has a multiple containing all ten digits.)

Checks

- Part 1 by hand: $13 \times 7 = 91$ and $29 \times 9 = 261$.
- A computer check confirmed the Part 1 recipe for every $n \leq 20\,000$ not ending in 5 or an even digit. It confirmed the Part 2 construction for every $n \leq 20\,000$ and $20\,000$

random n up to 10^{15}. It also computed the Part 3 multiplier for over 10 000 values of n of the required form, including ones ending in several zeros, and in every case $n \times m$ contained all ten digits. Examples: $21 \times 7\,201\,646\,090 = 151\,234\,567\,890$ and $11 \times 4\,657\,687\,990 = 51\,234\,567\,890$.

Common slips

- In Part 3, forgetting to check that the block 1234567890 really appears in full. The leading 1 needs $qm \geq 10^9$, which holds because $qm \equiv X$ and $qm > 0$.
- In Part 2, multiplying by 10^{a+b} instead of $2^b 5^a$. That adds zeros, but it leaves the 2s and 5s already in n where they are.

Beyond STEP

The pigeonhole argument in Part 3 proves a general fact: if $\gcd(q, N) = 1$ then q has a multiplicative inverse modulo N. The invertible residues form a group, written $(\mathbb{Z}/N\mathbb{Z})^\times$, and this is the starting point of elementary number theory and of RSA cryptography. A different trick shows that the final part holds for any n at all: some multiple of n lies between $X \cdot 10^k$ and $X \cdot 10^k + 10^k$ once $10^k \geq n$, and such a multiple *begins* with 1234567890. Davenport's *The Higher Arithmetic* is a lovely introduction to congruences and residues.

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